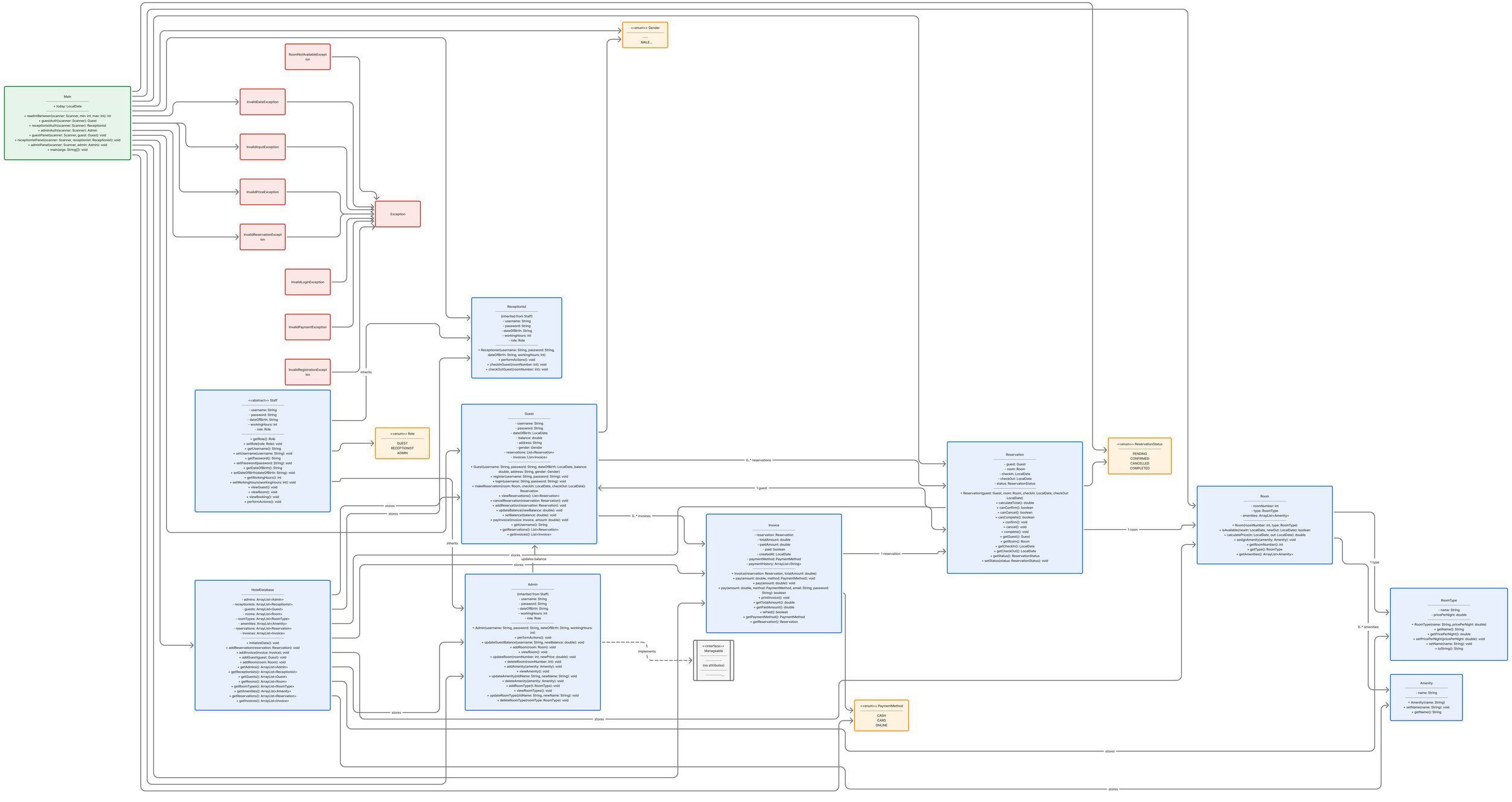

Main
+ today: LocalDate
+ readIntBetween(scanner: Scanner, min: int, max: int): int
+ guestAuth(scanner: Scanner): Guest
+ receptionistAuth(scanner: Scanner): Receptionist
+ adminAuth(scanner: Scanner): Admin
+ guestPanel(scanner: Scanner, guest: Guest): void
+ receptionistPanel(scanner: Scanner, receptionist: Receptionist): void
+ adminPanel(scanner: Scanner, admin: Admin): void
+ main(args: String[]): void
RoomNotAvailableException
InvalidDateException
InvalidInputException
InvalidPriceException
InvalidReservationException
InvalidLoginException
InvalidPaymentException
InvalidRegistrationException
Exception
inherits
<<enum>> Gender
___
MALE...
Receptionist
(inherited from Staff)
- username: String
- password: String
- dateOfBirth: String
- workingHours: int
- role: Role
+ Receptionist(username: String, password: String, dateOfBirth: String, workingHours: int)
+ performActions(): void
+ checkInGuest(roomNumber: int): void
+ checkOutGuest(roomNumber: int): void
<<abstract>> Staff
- username: String
- password: String
- dateOfBirth: String
- workingHours: int
- role: Role
+ getRole(): Role
+ setRole(role: Role): void
+ getUsername(): String
+ setUsername(username: String): void
+ getPassword(): String
+ setPassword(password: String): void
+ getDateOfBirth(): String
+ setDateOfBirth(dateOfBirth: String): void
+ getWorkingHours(): int
+ setWorkingHours(workingHours: int): void
+ viewGuest(): void
+ viewRoom(): void
+ viewBooking(): void
+ performActions(): void
<<enum>> Role
GUEST
RECEPTIONIST
ADMIN
Guest
- username: String
- password: String
- dateOfBirth: LocalDate
- balance: double
- address: String
- gender: Gender
- reservations: List<Reservation>
- invoices: List<Invoice>
+ Guest(username: String, password: String, dateOfBirth: LocalDate, balance: double, address: String, gender: Gender)
+ register(username: String, password: String): void
+ login(username: String, password: String): void
+ makeReservation(room: Room, checkIn: LocalDate, checkOut: LocalDate): Reservation
+ viewReservations(): List<Reservation>
+ cancelReservation(reservation: Reservation): void
+ addReservation(reservation: Reservation): void
+ updateBalance(newBalance: double): void
+ setBalance(balance: double): void
+ payInvoice(invoice: Invoice, amount: double): void
+ getUsername(): String
+ getReservations(): List<Reservation>
+ getInvoices(): List<Invoice>
0..* reservations
1 guest
0..* invoices
stores
stores
inherits
stores
updates balance
stores
Reservation
- guest: Guest
- room: Room
- checkIn: LocalDate
- checkOut: LocalDate
- status: ReservationStatus
+ Reservation(guest: Guest, room: Room, checkIn: LocalDate, checkOut: LocalDate)
+ calculateTotal(): double
+ canConfirm(): boolean
+ canCancel(): boolean
+ canComplete(): boolean
+ confirm(): void
+ cancel(): void
+ complete(): void
+ getGuest(): Guest
+ getRoom(): Room
+ getCheckIn(): LocalDate
+ getCheckOut(): LocalDate
+ getStatus(): ReservationStatus
+ setStatus(status: ReservationStatus): void
<<enum>> ReservationStatus
PENDING
CONFIRMED
CANCELLED
COMPLETED
1 room
Room
- roomNumber: int
- type: RoomType
- amenities: ArrayList<Amenity>
+ Room(roomNumber: int, type: RoomType)
+ isAvailable(newIn: LocalDate, newOut: LocalDate): boolean
+ calculatePrice(in: LocalDate, out: LocalDate): double
+ assignAmenity(amenity: Amenity): void
+ getRoomNumber(): int
+ getType(): RoomType
+ getAmenities(): ArrayList<Amenity>
1 type
0..* amenities
RoomType
- name: String
- pricePerNight: double
+ RoomType(name: String, pricePerNight: double)
+ getName(): String
+ getPricePerNight(): double
+ setPricePerNight(pricePerNight: double): void
+ setName(name: String): void
+ toString(): String
Amenity
- name: String
+ Amenity(name: String)
+ setName(name: String): void
+ getName(): String
Invoice
- reservation: Reservation
- totalAmount: double
- paidAmount: double
- paid: boolean
- createdAt: LocalDate
- paymentMethod: PaymentMethod
- paymentHistory: ArrayList<String>
+ Invoice(reservation: Reservation, totalAmount: double)
+ pay(amount: double, method: PaymentMethod): void
+ pay(amount: double): void
+ pay(amount: double, method: PaymentMethod, email: String, password: String): boolean
+ printInvoice(): void
+ getTotalAmount(): double
+ getPaidAmount(): double
+ isPaid(): boolean
+ getPaymentMethod(): PaymentMethod
+ getReservation(): Reservation
1 reservation
HotelDatabase
- admins: ArrayList<Admin>
- receptionists: ArrayList<Receptionist>
- guests: ArrayList<Guest>
- rooms: ArrayList<Room>
- roomTypes: ArrayList<RoomType>
- amenities: ArrayList<Amenity>
- reservations: ArrayList<Reservation>
- invoices: ArrayList<Invoice>
+ initializeData(): void
+ addReservation(reservation: Reservation): void
+ addInvoice(invoice: Invoice): void
+ addGuest(guest: Guest): void
+ addRoom(room: Room): void
+ getAdmins(): ArrayList<Admin>
+ getReceptionists(): ArrayList<Receptionist>
+ getGuests(): ArrayList<Guest>
+ getRooms(): ArrayList<Room>
+ getRoomTypes(): ArrayList<RoomType>
+ getAmenities(): ArrayList<Amenity>
+ getReservations(): ArrayList<Reservation>
+ getInvoices(): ArrayList<Invoice>
Admin
(inherited from Staff)
- username: String
- password: String
- dateOfBirth: String
- workingHours: int
- role: Role
+ Admin(username: String, password: String, dateOfBirth: String, workingHours: int)
+ performActions(): void
+ updateGuestBalance(username: String, newBalance: double): void
+ addRoom(room: Room): void
+ viewRoom(): void
+ updateRoom(roomNumber: int, newPrice: double): void
+ deleteRoom(roomNumber: int): void
+ addAmenity(amenity: Amenity): void
+ viewAmenity(): void
+ updateAmenity(oldName: String, newName: String): void
+ deleteAmenity(amenity: Amenity): void
+ addRoomType(t: RoomType): void
+ viewRoomTypes(): void
+ updateRoomType(oldName: String, newName: String): void
+ deleteRoomType(roomType: RoomType): void
implements
<<interface>> Manageable
(no attributes)
...
stores
stores
<<enum>> PaymentMethod
CASH
CARD
ONLINE
stores
stores